

Stochastic Processes

Lecture Notes & Solved Problems

Part I: Theoretical Foundations

A rigorous mathematical treatment of random processes, ergodicity, spectral analysis, Gaussian systems, Poisson models, and Markov chains.

PAGE 1: CONCEPTS AND BASIC TYPES

1. Random Processes: Concepts and Basic Types

• 1.1. Second-Order Processes

A random process $\{X(t), t \in T\}$ is a second-order process if its mean $E[X(t)]$ and variance $D[X(t)]$ exist for all $t \in T$. By the Cauchy-Schwarz inequality, the autocorrelation function $R_X(t, s) = E[X(t)X(s)]$ and autocovariance function $C_X(t, s) = E[(X(t) - E[X(t)])(X(s) - E[X(s)])]$ are guaranteed to exist.

• 1.2. Joint Autocorrelation and Cross-Correlation

The cross-correlation between two random processes $X(t)$ and $Y(t)$ is defined as $R_{XY}(t, s) = E[X(t)Y(s)]$. If they are jointly wide-sense stationary, it depends only on the time difference $\tau = t - s$:

$$R_{XY}(t, s) = R_{XY}(t - s)$$

Cross-correlation satisfies the symmetry property: $R_{XY}(\tau) = R_{YX}(-\tau)$. If the processes are real-valued, $R_{XY}(\tau) = R_{YX}(-\tau)$.

• 1.3. Stationary Processes (Strict-sense and Wide-sense)

- **Strict-Sense Stationary (SSS):** The joint distribution is invariant under any shift in time:

$$F_X(x_1, \dots, x_n; t_1, \dots, t_n) = F_X(x_1, \dots, x_n; t_1 + \tau, \dots, t_n + \tau)$$

- **Wide-Sense Stationary (WSS):** The mean is constant and the autocorrelation depends only on time difference:

$$E[X(t)] = \mu_X = \text{constant}, \quad R_X(t, s) = R_X(t - s)$$

• 1.4. Properties of the Autocorrelation Function

1. **Symmetry:** $R_X(\tau) = R_X(-\tau)$.
2. **Boundedness:** $|R_X(\tau)| \leq R_X(0) = E[X^2(t)]$.
3. **Relation to Autocovariance:**

$$C_X(t, s) = R_X(t, s) - E[X(t)]E[X(s)]$$

PAGE 2: MEAN ERGODICITY & KL EXPANSION

2. Mean Ergodicity

A WSS process $\{X(t)\}$ is mean-ergodic if its time average converges to its ensemble average in the mean-square sense as $T \rightarrow \infty$:

$$\hat{\mu}_T = \frac{1}{T} \int_{-T/2}^{T/2} X(t) dt \xrightarrow{\text{m.s.}} E[X(t)] = \mu_X$$

A necessary and sufficient condition for mean ergodicity is:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T \left(1 - \frac{|\tau|}{T}\right) C_X(\tau) d\tau = 0$$

A sufficient condition is that $\lim_{\tau \rightarrow \infty} C_X(\tau) = 0$.

3. Karhunen-Loève (KL) Expansion

The KL expansion represents a continuous-time second-order random process $X(t)$ over an interval $[a, b]$ as an infinite linear combination of orthogonal coordinate functions:

$$X(t) = \sum_{n=1}^{\infty} X_n \phi_n(t)$$

where $\{\phi_n(t)\}$ are deterministic orthonormal eigenfunctions satisfying the integral equation:

$$\int_a^b R_X(t, s) \phi_n(s) ds = \lambda_n \phi_n(t)$$

The coefficients X_n are uncorrelated random variables given by:

$$X_n = \int_a^b X(t) \phi_n^*(t) dt$$

and satisfy $E[X_n X_m^*] = \lambda_n \delta_{nm}$. The autocorrelation function can be expanded as:

$$R_X(t, s) = \sum_{n=1}^{\infty} \lambda_n \phi_n(t) \phi_n^*(s)$$

PAGE 3: LINEAR SYSTEMS AND SPECTRAL DENSITY

4. Linear Time-Invariant (LTI) Systems with Random Inputs

When a WSS process $X(t)$ is passed through a stable LTI system with impulse response $h(t)$ and transfer function $H(\omega)$, the output $Y(t) = h(t) * X(t)$ is also WSS.

- Output Mean:**

$$E[Y(t)] = E[X(t)] \int_{-\infty}^{\infty} h(t) dt = E[X(t)] H(0)$$

- Power Spectral Density (PSD) Relations:**

$$S_Y(\omega) = |H(\omega)|^2 S_X(\omega)$$

$$S_{YX}(\omega) = H(\omega)S_X(\omega), \quad S_{XY}(\omega) = \overline{H(\omega)}S_X(\omega)$$

5. Hilbert Transform and Analytic Signals

The Hilbert transform of a random process $X(t)$ is defined as:

$$\hat{X}(t) = X(t) * \frac{1}{\pi t} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{X(\tau)}{t - \tau} d\tau$$

The equivalent filter has transfer function $H(\omega) = -j \operatorname{sgn}(\omega)$.

- **Analytic Signal:** $Y(t) = X(t) + j\hat{X}(t)$.
- **Autocorrelation of Analytic Signal:**

$$R_Y(\tau) = 2R_X(\tau) + 2j\hat{R}_X(\tau)$$

- **PSD of Analytic Signal:**

$$S_Y(\omega) = 2S_X(\omega)[1 + \operatorname{sgn}(\omega)] = 4S_X(\omega)u(\omega)$$

PAGE 4: BANDPASS SIGNALS AND I/Q COMPONENTS

6. In-Phase and Quadrature (I/Q) Components of Bandpass Processes

A bandpass process $Y(t)$ centered at carrier frequency ω_c can be decomposed into in-phase $X_r(t)$ and quadrature $X_i(t)$ components:

$$Y(t) = X_r(t) \cos(\omega_c t) - X_i(t) \sin(\omega_c t)$$

$$\hat{Y}(t) = X_r(t) \sin(\omega_c t) + X_i(t) \cos(\omega_c t)$$

where $X_r(t)$ and $X_i(t)$ are lowpass processes defined as:

$$X_r(t) = Y(t) \cos(\omega_c t) + \hat{Y}(t) \sin(\omega_c t)$$

$$X_i(t) = \hat{Y}(t) \cos(\omega_c t) - Y(t) \sin(\omega_c t)$$

If $Y(t)$ is zero-mean and WSS, then $X_r(t)$ and $X_i(t)$ are jointly WSS with the following properties:

1. $E[X_r(t)] = E[X_i(t)] = 0$.
2. $R_{X_r}(\tau) = R_{X_i}(\tau) = R_Y(\tau) \cos(\omega_c \tau) + \hat{R}_Y(\tau) \sin(\omega_c \tau)$.
3. $R_{X_r X_i}(\tau) = R_Y(\tau) \sin(\omega_c \tau) - \hat{R}_Y(\tau) \cos(\omega_c \tau) = -R_{X_i X_r}(\tau)$.
4. $S_{X_r}(\omega) = S_{X_i}(\omega) = [S_Y(\omega - \omega_c) + S_Y(\omega + \omega_c)]_{LP}$.
5. $S_{X_r X_i}(\omega) = j[S_Y(\omega - \omega_c) - S_Y(\omega + \omega_c)]_{LP}$.

PAGE 5: GAUSSIAN PROCESSES

7. Gaussian Processes

1. Multivariate Density Function:

A vector of random variables $\mathbf{X} = (X_1, \dots, X_n)^T$ is Gaussian if its joint probability density is:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

where $\boldsymbol{\mu}$ is the mean vector and Σ is the positive-definite covariance matrix.

2. Characteristic Function:

$$\Phi_{\mathbf{X}}(\boldsymbol{\omega}) = E\{e^{j\boldsymbol{\omega}^T \mathbf{X}}\} = \exp\left(j\boldsymbol{\omega}^T \boldsymbol{\mu} - \frac{1}{2}\boldsymbol{\omega}^T \Sigma \boldsymbol{\omega}\right)$$

3. Moments property: A Gaussian process is completely specified by its first and second moments (mean and autocorrelation).

4. Linear Transformation: If $\mathbf{X}$ is Gaussian and $\mathbf{Y} = \mathbf{C}\mathbf{X} + \mathbf{d}$, then $\mathbf{Y}$ is also Gaussian.

5. Conditional Gaussian Distribution:

If $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)^T$ is Gaussian, the conditional distribution of $\mathbf{X}_1$ given $\mathbf{X}_2$ is also Gaussian:

$$\mathbf{X}_1 | \mathbf{X}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

6. Uncorrelatedness implies Independence: For Gaussian variables, they are independent if and only if they are uncorrelated ($\Sigma_{12} = 0$).

PAGE 6: GAUSS-MARKOV & POISSON PROCESSES

8. Gauss-Markov Processes

A Gaussian process $X(t)$ is Markov if and only if for any $t_1 < t_2 < t_3$, the correlation satisfies the relation:

$$C_X(t_1, t_3) = \frac{C_X(t_1, t_2)C_X(t_2, t_3)}{C_X(t_2, t_2)}$$

which implies the covariance factorizes as $C_X(t, s) = h(\min(t, s))g(\max(t, s))$.

9. Poisson Processes

A Poisson process $\{N(t), t \geq 0\}$ with intensity $\lambda > 0$ is a discrete-state, continuous-time process representing count statistics of independent events:

$$P(N(t) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, \quad n = 0, 1, \dots$$

- **Mean and Variance:** $E[N(t)] = D[N(t)] = \lambda t$.
- **Autocorrelation Function:**

$$R_N(t, s) = E[N(t)N(s)] = \lambda^2 ts + \lambda \min(t, s)$$

- **Autocovariance Function:**

$$C_N(t, s) = \lambda \min(t, s)$$

PAGE 7: ARRIVAL TIMES AND ORDER STATISTICS

10. Poisson Process Arrival Times and Waiting Times

1. Inter-Arrival Times (T_n):

The time intervals between consecutive events $T_n = S_n - S_{n-1}$ are independent, identically distributed (i.i.d.) exponential random variables:

$$f_{T_n}(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \quad E[T_n] = \frac{1}{\lambda}, \quad D[T_n] = \frac{1}{\lambda^2}$$

2. Waiting Times (S_n):

The arrival time of the n -th event $S_n = \sum_{i=1}^n T_i$ follows a Gamma (Erlang) distribution:

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1}}{(n-1)!} e^{-\lambda t}, \quad t \geq 0$$

3. Conditional Distribution of Arrival Times:

Given that $N(t) = n$ events occurred in the interval $[0, t]$, the arrival times $\{S_1, \dots, S_n\}$ are distributed as the order statistics of n independent random variables uniformly distributed on $[0, t]$:

$$f_{S_1, \dots, S_n | N(t)=n}(u_1, \dots, u_n) = \frac{n!}{t^n}, \quad 0 \leq u_1 \leq \dots \leq u_n \leq t$$

PAGE 8: COMPOUND & NON-HOMOGENEOUS POISSON PROCESSES

11. Compound Poisson Processes

A compound Poisson process $\{Y(t), t \geq 0\}$ is defined as $Y(t) = \sum_{k=1}^{N(t)} H_k$, where $\{N(t)\}$ is a Poisson process and $\{H_k\}$ are i.i.d. random variables independent of $N(t)$.

- **Mean:** $E[Y(t)] = \lambda t E[H_k]$.
- **Variance:**

$$D[Y(t)] = \lambda t E[H_k^2]$$

- **Characteristic Function:**

$$G_Y(u) = e^{\lambda t [G_H(u) - 1]}$$

12. Non-Homogeneous Poisson Processes

If the rate of arrival varies with time as $\lambda(t)$, the process has independent increments, and the count follows:

$$P(N(t) - N(s) = n) = \frac{\Lambda(s, t)^n}{n!} e^{-\Lambda(s, t)}, \quad \text{where } \Lambda(s, t) = \int_s^t \lambda(u) du$$

PAGE 9: MARKOV CHAINS

13. Discrete-Time Markov Chains (DTMC)

1. Markov Property:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i) = P_{ij}$$

2. Chapman-Kolmogorov Equations:

$$P_{ij}^{(m+n)} = \sum_k P_{ik}^{(m)} P_{kj}^{(n)}$$

3. State Classification:

- **Recurrent:** A state is recurrent if the probability of eventually returning to it is 1 ($f_{ii} = 1$). - If mean return time $\mu_i = E[T_{ii}] < \infty$, the state is **positive recurrent**. - If $\mu_i = \infty$, it is **null recurrent**.
- **Transient:** A state is transient if $f_{ii} < 1$, meaning $\sum_{n=1}^{\infty} P_{ii}^{(n)} < \infty$.

4. Stationary Distribution (boldsymbol{\pi}):

For an irreducible, aperiodic Markov chain, the stationary distribution satisfies the balance equations:

$$\pi_j = \sum_k \pi_k P_{kj}, \quad \sum_j \pi_j = 1$$

The limiting distribution is given by $P_{ij}^{(n)} \rightarrow \pi_j = 1/\mu_j$ as $n \rightarrow \infty$.

PAGE 10: AUTOCORRELATION-PSD TRANSFORMS

14. Common Autocorrelation and Power Spectral Density Pairs

Autocorrelation Function $R_X(\tau)$	Power Spectral Density $S_X(\omega)$
$e^{-\alpha \tau }$	$\frac{2\alpha}{\omega^2 + \alpha^2}$
$\delta(\tau)$	1
$\cos(\omega_0\tau)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\frac{\sin(\omega_0\tau)}{\pi\tau}$	$\text{rect}(\frac{\omega}{2\omega_0})$
$e^{-\alpha\tau^2}$	$\sqrt{\frac{\pi}{\alpha}}e^{-\frac{\omega^2}{4\alpha}}$

Part II: Solved Applied Problems

Detailed, step-by-step mathematical proofs and solutions to exam and homework problems on WSS processes, LTI transformations, Gaussian systems, Poisson processes, and expectations.

PAGE 11: PROBLEMS ON WIDE-SENSE STATIONARITY

Problem 1: Let $X(t) = f(t + \eta)$, where $f(t)$ is a deterministic periodic function with period T , and η is a random variable uniformly distributed on $[0, T]$. Prove that $X(t)$ is wide-sense stationary.

Proof:

1. Compute the mean:

$$E[X(t)] = E[f(t + \eta)] = \frac{1}{T} \int_0^T f(t + u) du$$

Let $v = t + u$, then $du = dv$. Since $f(v)$ is periodic with period T , the integral over one full period is constant and independent of the shift t :

$$E[X(t)] = \frac{1}{T} \int_t^{t+T} f(v) dv = \frac{1}{T} \int_0^T f(v) dv = \mu_X = \text{constant}$$

2. Compute the autocorrelation:

$$R_X(t, s) = E[X(t)X(s)] = E[f(t + \eta)f(s + \eta)] = \frac{1}{T} \int_0^T f(t + u)f(s + u) du$$

Let $v = t + u$, then:

$$R_X(t, s) = \frac{1}{T} \int_t^{t+T} f(v)f(v + s - t) dv$$

Using the periodicity of f , this integral is independent of t and depends only on $\tau = t - s$:

$$R_X(t, s) = \frac{1}{T} \int_0^T f(v)f(v - \tau) dv = R_X(\tau)$$

Since the mean is constant and the autocorrelation depends only on the time difference, $X(t)$ is WSS.

PAGE 12: DERIVATIVES AND INTEGRALS OF WSS PROCESSES

Problem 2: Let $X(t)$ be a WSS process with autocorrelation function $R_X(\tau) = e^{-\tau^2}$. Let $Y(t) = X(t) + X'(t)$. Compute the cross-correlation $R_{XY}(t, s)$ and the autocorrelation $R_Y(t, s)$.

Solution:

1. Compute the cross-correlation:

$$R_{XY}(t, s) = E[X(t)Y(s)] = E[X(t)(X(s) + X'(s))] = E[X(t)X(s)] + E[X(t)X'(s)]$$

Using properties of derivatives of stationary processes:

$$R_{XY}(t, s) = R_X(t - s) + \frac{\partial}{\partial s} R_X(t - s) = R_X(\tau) - R_X'(\tau)$$

For $R_X(\tau) = e^{-\tau^2}$, we have $R_X'(\tau) = -2\tau e^{-\tau^2}$, so:

$$R_{XY}(\tau) = (1 + 2\tau)e^{-\tau^2}$$

2. Compute the autocorrelation of $Y(t)$:

$$R_Y(t, s) = E[Y(t)Y(s)] = E[(X(t) + X'(t))(X(s) + X'(s))]$$

$$R_Y(t, s) = E[X(t)X(s)] + E[X(t)X'(s)] + E[X'(t)X(s)] + E[X'(t)X'(s)]$$

$$R_Y(\tau) = R_X(\tau) + R_X'(\tau) - R_X'(\tau) - R_X''(\tau) = R_X(\tau) - R_X''(\tau)$$

Differentiating $R'_X(\tau) = -2\tau e^{-\tau^2}$:

$$R''_X(\tau) = -2e^{-\tau^2} + 4\tau^2 e^{-\tau^2}$$

Thus:

$$R_Y(\tau) = e^{-\tau^2} - (-2e^{-\tau^2} + 4\tau^2 e^{-\tau^2}) = (3 - 4\tau^2)e^{-\tau^2}$$

PAGE 13: LINEAR SYSTEMS AND FILTERING PROBLEMS

Problem 3: Let $X(t)$ be a WSS process passing through an LTI system with transfer function $H(\omega)$. Let $Z(t) = Y(t) - X(t)$ be the output tracking error, where $Y(t) = H(\omega) * X(t)$. Show that the PSD of the error is:

$$S_Z(\omega) = |H(\omega) - 1|^2 S_X(\omega)$$

Proof:

1. Write the autocorrelation of $Z(t)$:

$$R_Z(\tau) = E[Z(t + \tau)Z(t)] = E[(Y(t + \tau) - X(t + \tau))(Y(t) - X(t))]$$

$$R_Z(\tau) = R_Y(\tau) - R_{YX}(\tau) - R_{XY}(\tau) + R_X(\tau)$$

2. Convert to the frequency domain using the Fourier Transform:

$$S_Z(\omega) = S_Y(\omega) - S_{YX}(\omega) - S_{XY}(\omega) + S_X(\omega)$$

Recall LTI relations: $S_Y(\omega) = |H(\omega)|^2 S_X(\omega)$, $S_{YX}(\omega) = H(\omega) S_X(\omega)$, and $S_{XY}(\omega) = \overline{H(\omega)} S_X(\omega)$.

$$S_Z(\omega) = [|H(\omega)|^2 - H(\omega) - \overline{H(\omega)} + 1] S_X(\omega)$$

Factorizing the term inside the brackets:

$$S_Z(\omega) = (H(\omega) - 1)(\overline{H(\omega)} - 1) S_X(\omega) = |H(\omega) - 1|^2 S_X(\omega)$$

This completes the proof.

PAGE 14: SPECTRAL FACTORIZATION

Problem 4: Given a system power spectral response $|H(\omega)|^2 = \frac{\omega^2 + 8}{\omega^2 + 3}$, factorize it to find a stable and minimum-phase transfer function $H(s)$.

Solution:

1. Substitute $s = j\omega$ (so $\omega^2 = -s^2$):

$$H(s)H(-s) = \frac{-s^2 + 8}{-s^2 + 3} = \frac{s^2 - 8}{s^2 - 3} = \frac{(s - \sqrt{8})(s + \sqrt{8})}{(s - \sqrt{3})(s + \sqrt{3})}$$

2. Group the poles and zeros to construct a stable and minimum-phase transfer function. A system is stable if its poles lie in the left half of the s-plane ($\text{Re}(s) < 0$). It is minimum-phase if its zeros also lie in the left half of the s-plane ($\text{Re}(s) < 0$). Thus, we choose:

$$H(s) = \frac{s + \sqrt{8}}{s + \sqrt{3}} \Rightarrow H(\omega) = \frac{j\omega + \sqrt{8}}{j\omega + \sqrt{3}}$$

PAGE 15: JOINT GAUSSIANTY AND CONDITIONAL EXPECTATIONS

Problem 5: Let $X(t)$ be a Gaussian process with PSD $S_X(\omega) = \frac{2\sigma^2}{\omega^2/\alpha^2 + 1}$. It is passed through a stable filter $H(s) = \frac{1}{s + \beta}$. Find the PSD $S_Y(\omega)$ and the autocorrelation $R_Y(\tau)$ of the output process $Y(t)$.

Solution:

1. Determine $S_Y(\omega)$:

$$S_Y(\omega) = |H(\omega)|^2 S_X(\omega) = \frac{1}{\omega^2 + \beta^2} \cdot \frac{2\sigma^2\alpha^2}{\omega^2 + \alpha^2}$$

2. Apply partial fraction expansion:

$$S_Y(\omega) = \frac{2\sigma^2\alpha^2}{\alpha^2 - \beta^2} \left[\frac{1}{\omega^2 + \beta^2} - \frac{1}{\omega^2 + \alpha^2} \right]$$

3. Take the inverse Fourier Transform to find $R_Y(\tau)$:

$$R_Y(\tau) = \frac{\sigma^2\alpha^2}{\alpha^2 - \beta^2} \left[\frac{e^{-\beta|\tau|}}{\beta} - \frac{e^{-\alpha|\tau|}}{\alpha} \right]$$
